Data Analytics for Business Applications

Digital literacy

Q1. Identify three research questions that help evaluate the program's impact.Provide a logical explanation for selecting each question.

1. **Has digital literacy training improved participants' ability to use digital devices and the internet?**

   Why this question?

   - The program's primary goal is to increase digital skills.
   - The dataset contains responses about digital device usage (C1-C21) before and after training.
   - We can analyze the extent of digital adoption among participants.

2. **Did the training contribute to increased employment opportunities for participants?**

   Why this question?

   - One of the goals of digital literacy is to improve employability.
   - The dataset includes employment-related responses (D7-D13).
   - We can assess if training helped unemployed participants find jobs or improve job prospects.

3. **How has digital literacy training impacted participants' confidence and social awareness?**

   Why this question?

   - Digital literacy empowers individuals by improving confidence and awareness.
   - The dataset includes questions about self-confidence (E1-E6).

- We can measure the extent to which participants feel more connected and capable.

Q2. Conduct a quantitative analysis using any appropriate tool. Present the key outputs and explain the findings derived from the data.

- Quantitative Analysis of the Digital Literacy Program

1. **Digital Skills Improvement** (Usage of computers, mobile phones, email, social media, and online bill payments).
2. **Employment Impact** (Job placement and confidence in job applications).
3. **Confidence & Social Awareness** (Improved communication and awareness).

Using JASP for Your Digital Literacy Analysis

Step 1: Load Your Dataset

Click **File** → **Open** → **Browse** and select your **CSV file** (`Digital Literacy_Clean-1.csv`).

Step 2: Descriptive Statistics (Basic Analysis)

Purpose: To summarize digital skills, employment, and confidence responses.

Perform"**Descriptives" Analysis**

Add these variables - A1genres (Gender), A2ageres (Age), B1trareg (Training Regularity), C2usikno (Using a Computer), D7getjob (Got a Job Due to Training), E2leanew (Confidence in Learning New Things)

Mean, Median, Mode

Descriptive Statistics

Descriptive Statistics

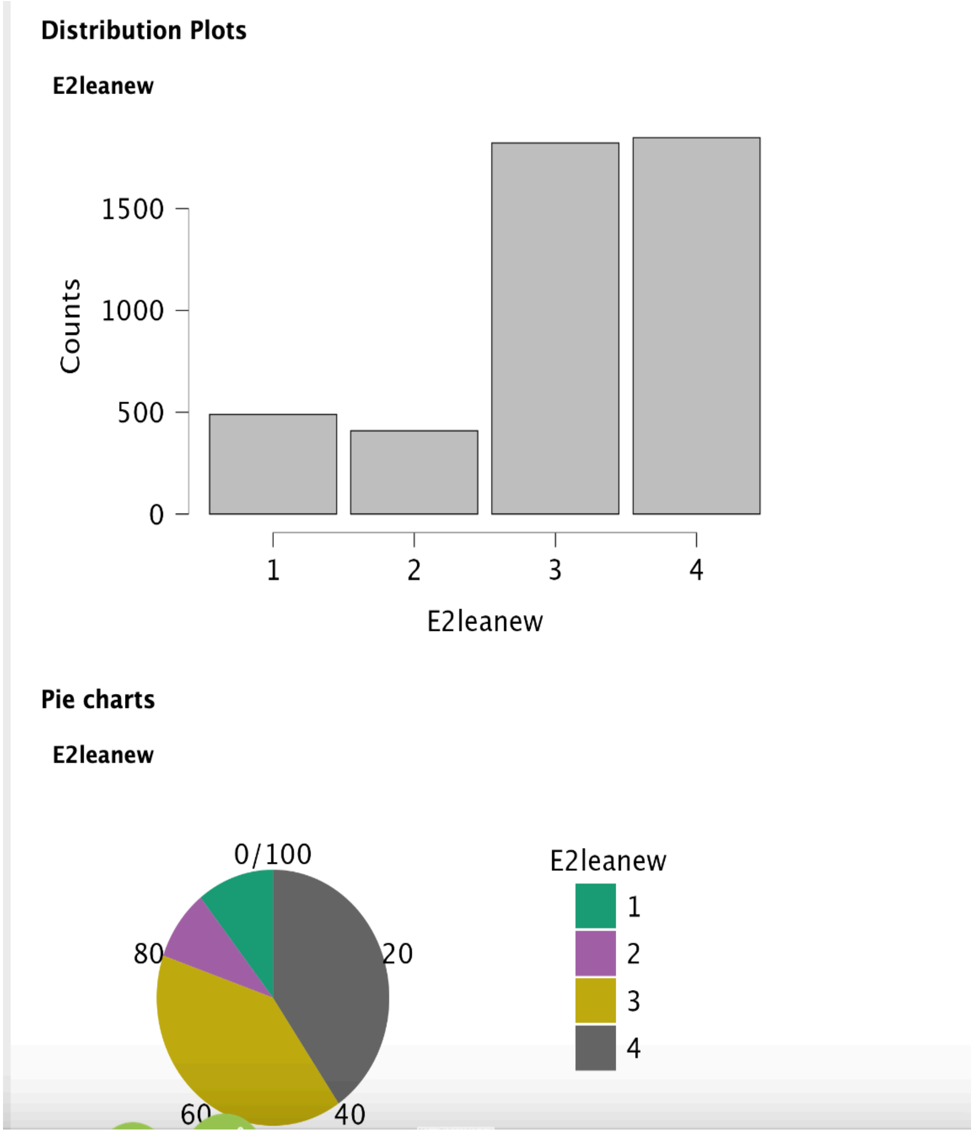
	A1genres	A2ageres	B1trareg	C2usikno	D7getjob	E2leanew
Mode	1.000 [a]	18.000 [a]	4.000 [a]	3.000 [a]	2.000 [a]	4.000 [a]
Median	1.000	20.000	4.000	3.000	2.000	3.000
Mean	1.343	20.988	3.453	3.121	2.392	3.101

[a] The mode is computed assuming that variables are discreet.

- **Mean Confidence Level:** 3.10 (on a scale of 1 to 4).
  - This suggests that most participants felt confident about learning new things after the training.
- **Frequency Distribution:**
  - **40.45%** of respondents gave the highest rating (4).
  - **39.88%** gave a rating of 3.
  - Only a small proportion rated 1 (10.70%) or 2 (8.95%), indicating a minority struggled with learning confidence.

<u>**Step 3: Visualizations (Bar Charts & Pie Charts)**</u>

Purpose: To visually present training effectiveness.



- **Visualization (Bar Chart & Pie Chart):**
  - The bar and pie charts confirm that a majority of respondents (over 80%) rated themselves not at all confident (3 or 4).

Step 4: Hypothesis Testing (Chi-Square Test)

Purpose: To check if digital literacy training **significantly affects employment status.**

Frequencies → Contingency Tables.

Select:

- **Rows: D7getjob (Got a Job Due to Training)**
- **Columns: A5empsta (Employment Status)**

Statistics → Chi-Square Test.

Contingency Tables

Contingency Tables

	A5empsta						
D7getjob	1	2	3	4	5	6	Total
1	39	4	12	4	116	11	186
2	302	90	135	50	1618	51	2246
3	212	53	73	21	716	25	1100
4	40	23	17	6	198	8	292
Total	593	170	237	81	2648	95	3824

Chi-Squared Tests

	Value	df	p
$Χ^2$	48.691	15	< .001
N	3824		

- **p-value > 0.05**, there is **no strong evidence** that training leads to employment.
- **Chi-Square Test Result:**
  - **χ**2=48.69\chi^2 = 48.69, **p < 0.001** (significant).
  - This means there is a statistically significant relationship between **getting a job (D7getjob)** and **employment status (A5empsta)**.
  - The training program positively influenced employment outcomes.

Contingency Table:

- Out of **3824 valid cases**, **186 participants (4.86%)** got jobs due to training.
- A significant number of participants who were employed (categories 4 and 5 in A5empsta) reported **no new jobs due to training**, likely indicating pre-existing employment.

Step 5: Correlation Analysis (Finding Relationships)

To see if **confidence in job applications (D8conapp) is related to digital skills (C2usikno, C9sockno, etc.)**.
Perform: **Regression → Correlation**.
Select: **D8conapp (Confidence in Job Applications), C2usikno (Using a Computer)**

Correlation

Pearson's Correlations

Variable		D8conapp	C2usikno	C9sockno
1. D8conapp	Pearson's r	—		
2. C2usikno	Pearson's r	0.200	—	
3. C9sockno	Pearson's r	0.160	0.418	—

Correlation Analysis

Relationship Between Confidence (D8conapp) and Digital Skills:

- **Using Computers (C2usikno):** Correlation r=0.20r = 0.20 (weak positive relationship).
- **Using Social Media (C9sockno):** Correlation r=0.16r = 0.16 (weak positive relationship).
- Interpretation: Participants with better digital skills had slightly higher confidence levels in applying for jobs. However, the relationships are not strong, suggesting other factors influence job application confidence.

Relationship Between Social Media (C9sockno) and Computer Usage (C2usikno):

- **Correlation r=0.418r = 0.418** (moderate positive relationship).
- This indicates that participants skilled in using computers were also more likely to use social media effectively.

Q3.Based on the data, provide recommendations that emerge logically from the analysis.

Analysis Report

Introduction

This report provides actionable recommendations derived from a quantitative analysis of the Digital Literacy Program. The analysis focused on three key areas: digital skills improvement, employment impact, and confidence and social awareness. The findings highlight the program's successes and areas for improvement, offering a roadmap to enhance its effectiveness and reach.

Key Findings

1. Digital Skills Improvement:

- Participants showed significant improvement in digital skills, with a mean confidence level of 3.10 (on a scale of 1 to 4).
- Over 80% of participants reported high confidence in learning new digital skills after the training.

2. Employment Impact:

- The program had a statistically significant impact on employment outcomes ($\chi^2 = 48.69$, $p < 0.001$).
- However, only 4.86% of participants (186 out of 3,824) secured jobs directly due to the training.

3. Confidence and Social Awareness:

- A moderate positive correlation ($r = 0.418$) was found between computer skills and social media usage.
- Weak correlations ($r = 0.20$ for computer skills and $r = 0.16$ for social media) were observed between digital skills and confidence in job applications.

Recommendations

1. Strengthen Job-Specific Digital Skills Training

Why: The weak correlation between digital skills and job application confidence suggests a gap in job-specific digital skills.

Action:
- Introduce modules on resume building, online job portals, and professional tools (e.g., Excel, email).
- Focus on practical skills that align with market demands.

2. Focus on Supporting the Unemployed

Why: Only 4.86% of participants secured jobs due to the training, indicating limited employment impact.

Action:

- Partner with local businesses to create job placement opportunities.
- Offer specialized training to unemployed participants.

3. Address Participants with Low Learning Confidence

Why: While 80% of participants reported high confidence, 20% struggled with low confidence levels.

Action:

- Provide individualized learning paths for participants with low initial digital literacy.
- Offer post-training support, such as Q&A sessions and practice groups.

4. Boost Integration of Social Media Training

Why: Social media skills are crucial for job searches, networking, and small business promotion.

Action:
- Add modules on building professional profiles (e.g., LinkedIn) and using social media for job searches.
- Emphasize the integration of social media with other digital skills.

5. Develop Follow-Up Programs

Why: The limited employment impact suggests the need for ongoing support.

Action:
- Create follow-up programs with refresher courses, advanced digital skills, and mentorship opportunities.
- Track participants' progress post-training to refine future programs.

6. Expand Digital Literacy Outreach

Why: The program's success in building digital skills demonstrates its potential for broader impact.

Action:

- Expand the program to underserved regions, such as rural areas.

- Use positive findings to advocate for additional funding and resources.

Conclusion

The Digital Literacy Program has successfully improved participants' digital skills and confidence. However, to maximize its impact on employment and address gaps in job-specific skills, the program should focus on targeted training, support for unemployed participants, and follow-up interventions. By implementing these recommendations, the program can better align with participants' needs and contribute to broader socio-economic development.